NETFLIX ANALYTICS DASHBOARD REPORT

1. ABSTRACT

This project presents a Netflix analytics dashboard developed using Python and Power BI to analyze user behavior, revenue trends, and content preferences. The dashboard integrates multiple KPIs and visualizations to provide a unified and interactive view of platform performance.

KPI dashboards are designed to give a consolidated view of business performance and enable faster decision-making by presenting key metrics in a single interface.

2. PROBLEM STATEMENT

Streaming platforms generate large volumes of user data, but raw datasets do not directly provide meaningful insights. Key challenges include:

- Difficulty in identifying revenue drivers
- Lack of clarity in user engagement patterns
- No structured view of content popularity

The objective is to transform raw data into a structured analytical dashboard that supports business decisions.

3. OBJECTIVES

- Analyze user engagement behavior
- Identify high-performing genres
- Evaluate subscription-based revenue contribution
- Measure binge-watching behavior
- Track usage trends over time
- Enable interactive filtering for exploratory analysis

4. DATA PROCESSING AND TRANSFORMATION

4.1 Data Cleaning

- Removed duplicate records
- Handled missing values
- Standardized column formats

4.2 Feature Engineering

From the dataset:

- Converted watch time from minutes to hours
- Created engagement categories:
  - Low (<1 hour)
  - Medium (1–3 hours)
  - High (>3 hours)

4.3 User-Level Aggregation

Generated a user-level dataset using:

- Total watch hours per user
- Average rating per user
- Average monthly fee
- Binge rate per user

This transformation improves analytical accuracy by shifting from transaction-level data to behavioral insights.

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5. KPI FRAMEWORK

A Key Performance Indicator (KPI) is a measurable value that reflects how effectively a system achieves its objectives .

Selected KPIs

KPI	Description	Purpose
Total Users	Number of unique users	Measures platform scale
Total Revenue	Sum of subscription fees	Indicates financial performance
Average Rating	Mean user rating	Reflects content quality
Binge Rate (%)	% of binge users	Measures engagement intensity
Total Watch Hours	Aggregate consumption	Indicates usage behavior

These KPIs provide both **operational** and **strategic insights**.

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6. DASHBOARD DESIGN AND VISUALIZATION

6.1 Revenue Analysis

- Visualization: Bar Chart

- Metric: Revenue by subscription type

Insight:
Standard subscription contributes the highest share of revenue, indicating optimal pricing balance.

6.2 Genre Distribution

- Visualization: Donut Chart
- Metric: User count by genre

Insight:
Drama dominates user preference, suggesting strong audience demand.

6.3 Binge Behavior Analysis

- Visualization: Pie Chart
- Metric: Binge vs non-binge users

Insight:
A significant proportion of users exhibit binge-watching behavior, indicating high engagement potential.

6.4 Engagement Level Distribution

- Visualization: Bar Chart
- Metric: Users by engagement category

Insight:
Most users fall into the medium engagement segment, representing an opportunity for conversion to high engagement.

6.5 Trend Analysis

- Visualization: Line Chart
- Metric: Watch hours over quarters

Insight:
An increasing trend indicates growing platform usage and retention.

6.6 Genre Popularity (Replaced Chart)

- Visualization: Horizontal Bar Chart
- Metric: Count of users by genre

Purpose:
Replaces less meaningful metrics (such as age-based analysis) with direct business insights.

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7. ANALYTICAL INSIGHTS

7.1 Revenue Optimization

- Standard plans drive maximum revenue
- Premium plans have lower adoption but higher individual value

Implication:
Focus on converting basic users to standard plans.

7.2 Content Strategy

- Drama is the most consumed genre
- Other genres show comparatively lower engagement

Implication:
Increase investment in high-performing genres while improving visibility of others.

7.3 User Engagement

- Medium engagement users form the majority
- High engagement users are fewer but highly valuable

Implication:
Target medium users with personalized recommendations.

7.4 Behavioral Patterns

- Binge watchers represent a critical segment

Implication:
Introduce features such as autoplay and recommendations to enhance retention.

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8. DESIGN PRINCIPLES

The dashboard follows standard BI design principles:

- Centralized KPI display for quick overview
- Minimal and meaningful visuals
- Consistent color theme and layout
- Interactive filters for exploration

Effective dashboards combine multiple metrics into a single interface to provide a unified view of performance and support decision-making .

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9. LIMITATIONS

- Static dataset (no real-time updates)
- No predictive analytics
- Limited segmentation depth
- Basic feature engineering

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10. FUTURE ENHANCEMENTS

- Integration of machine learning models for prediction
- User churn analysis
- Recommendation system implementation
- Real-time data pipeline integration
- Advanced segmentation (behavioral clustering)

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11. CONCLUSION

The Netflix analytics dashboard successfully transforms raw data into structured insights using Power BI. It enables analysis of user behavior, revenue patterns, and content trends in a clear and interactive format.

The project demonstrates the practical application of data analytics in supporting business decisions and improving user engagement strategies.